

Naveed Shaik

naveedshaik120@gmail.com | +1 940-808-9188 | [LinkedIn](#) | [GitHub](#) | [portfolio](#) | United States

PROFESSIONAL SUMMARY

AI and Data Engineer with 4+ years of experience designing scalable data platforms and building production-grade AI solutions across banking and healthcare domains. Experienced in developing **Agentic AI**, **Retrieval-Augmented Generation (RAG)**, **LLM-powered applications**, and cloud-native **data pipelines** using Azure, AWS, Databricks, PySpark, and GPT-4 to solve complex business challenges. Skilled at translating business requirements into secure, high-performance AI and data engineering solutions while optimizing reliability, observability, and operational efficiency. Passionate about leveraging modern AI and data technologies to help organizations accelerate innovation, improve decision-making, and deliver measurable business value.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, HTML, CSS, React, PySpark

AI Engineering & GenAI: Azure OpenAI **GPT-4**, LangChain, LangGraph, Agentic AI, Multi-Agent Systems, Supervisor Agent, Retrieval-Augmented Generation (RAG), Auto-RAG, Cache-based RAG, MCP, CrewAI, ReAct, Chain-of-Thought (CoT), Advanced Prompt Engineering, PydanticAI, FastAPI, Uvicorn

LLM Evaluation & Observability: Ragas, DeepEval, Faithfulness, Answer Relevancy, Context Precision, Context Recall, Prompt Optimization, Model Monitoring, Prometheus, Grafana

Data Engineering: ETL Pipelines, Data Pipelines, Apache Spark, PySpark, Databricks, Medallion Architecture (Bronze/Silver/Gold), Apache Airflow, MLflow, Hadoop, Hive, Impala, JSON, XML, Data Validation, Data Quality, Data Transformation

Cloud & DevOps: Microsoft Azure, Azure OpenAI, Azure Kubernetes Service (AKS), Kubernetes, RBAC, AWS (S3, EC2, SageMaker, Bedrock), GCP, Vertex AI, Docker, CI/CD

Databases & Vector Stores: PostgreSQL, MySQL, MongoDB, Snowflake, Redis, Pinecone, ChromaDB, FAISS, BigQuery

Machine Learning Operations: Feature Engineering, Fine-tuning (LoRA, QLoRA), Model Deployment, Inference Optimization, MLflow

Visualization & Analytics: Power BI, Tableau, Excel, Matplotlib

Version Control & Collaboration: Git, Jira, Agile/Scrum, RESTful APIs

PROFESSIONAL EXPERIENCE

AI Engineer | Wells Fargo | July 2024 – Present

- Architected an **Agentic AI system** using **Azure OpenAI GPT-4**, **LLMs**, **LangChain**, **LangGraph**, and **Retrieval-Augmented Generation (RAG)**, leveraging **Chain-of-Thought (CoT)** and **ReAct** reasoning patterns to enable intelligent multi-step reasoning, improving AI task completion accuracy by **38%**.
- Engineered a **Supervisor Agent** to intelligently decompose complex user requests and orchestrate specialized sub-agents, reducing average task execution time by **32%** while improving workflow scalability across enterprise banking use cases.
- Designed a **dual-layer memory architecture** by implementing **cache-based RAG** for low-latency conversational context and long-term semantic retrieval over enterprise knowledge repositories, reducing response latency by **45%** and improving response consistency.
- Developed **Auto-RAG** pipelines that dynamically validated user queries and retrieved banking policies, compliance documents, and knowledge articles only when required, reducing hallucinations by **30%** and lowering unnecessary retrieval operations by **35%**.
- Applied **advanced prompt engineering techniques**, including role prompting, few-shot prompting, **Chain-of-Thought (CoT)**, **ReAct**, and structured output prompting to translate complex banking business rules into production-grade AI agent behavior, increasing response relevance by **28%**.
- Established a comprehensive LLM evaluation framework using **Ragas**, **DeepEval**, and quality metrics including **Faithfulness**, **Answer Relevancy**, **Context Precision**, **Context Recall**, and latency metrics, achieving **94% Faithfulness** and **92% Answer Relevancy** across production evaluation datasets.
- Built automated evaluation pipelines using Ragas and DeepEval with curated banking question-answer datasets to benchmark retrieval quality, hallucination rates, and response consistency, reducing manual validation effort by 60%.
- Designed agent evaluation metrics including task completion rate, tool selection accuracy, reasoning quality, and latency to continuously optimize multi-agent workflows and improve end-to-end execution reliability by 20%.
- Deployed and supported enterprise AI applications on **Microsoft Azure** using **Azure Kubernetes**, implementing end-to-end observability with **Prometheus** and **Grafana** to monitor model latency, token consumption, retrieval quality, API health, and production system performance, maintaining **99.9% service availability**.

- Partnered with product owners, compliance teams, and business stakeholders to deliver secure, production-ready AI solutions for banking workflows, ensuring responsible AI practices, governance, auditability, and regulatory compliance while reducing manual support effort by **40%** through continuous prompt refinement and retrieval optimization.

Data Engineer | Accenture | Jan 2022 – Dec 2023

- Engineered and maintained scalable **PySpark** data pipelines on **Databricks** and **AWS** to ingest, transform, and process **electronic health record (EHR)**, patient, claims, and clinical data for enterprise analytics and reporting.
- Designed and implemented a **Medallion Architecture (Bronze, Silver, Gold)** in **Databricks**, standardizing healthcare data processing and delivering trusted datasets for business intelligence, regulatory reporting, and operational analytics.
- Developed and optimized complex **SQL** and **PostgreSQL** queries to transform, validate, and aggregate large healthcare datasets, improving data accessibility for analysts, clinicians, and business stakeholders.
- Built reusable **PySpark ETL** workflows to cleanse, standardize, and integrate structured and semi-structured healthcare data while implementing data quality validations to ensure accurate and consistent reporting.
- Collaborated with data scientists to integrate and support **machine learning models** for patient risk prediction and healthcare analytics, monitoring model performance using **Precision (92%)**, **Recall (89%)**, **F1-Score (90%)**, and **ROC-AUC (0.94)** to maintain reliable predictive outcomes.
- Partnered with cross-functional teams to automate data ingestion, troubleshoot production pipeline issues, and optimize cloud-based data workflows on **AWS**, ensuring secure, reliable, and scalable data delivery for downstream analytics and reporting.

PERSONAL PROJECT

Intelligent Payment Operations & Transaction Insights Platform

- Developed an event-driven transaction intelligence platform using **Python**, **FastAPI**, **Apache Kafka**, **PostgreSQL**, and **Redis** to process and analyze over **750K simulated payment events**, reducing end-to-end processing latency by **38%** through asynchronous message handling and distributed consumers.
- Built AI-assisted transaction analysis services that combined **LLM-powered summaries** with rule-based validation to categorize payment anomalies, generate operational insights, and improve transaction review accuracy by **27%**, while implementing structured JSON schema validation for reliable downstream processing.
- Designed scalable microservices with **REST APIs**, **AsyncIO**, **Docker**, and **Kubernetes**, incorporating retry mechanisms, idempotent event processing, centralized logging, and health monitoring, achieving **99.8% service availability** during high-volume workload simulations.
- Implemented an evaluation and observability pipeline using **Prometheus**, **Grafana**, and **automated regression tests**, tracking API latency, processing throughput, and model quality metrics, reducing production issue detection time by **45%** and improving deployment confidence through continuous performance monitoring.

FinAgent AI — RAG-Powered Banking Document Intelligence Platform

- Designed and implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **ChromaDB** embeddings and a locally-hosted **Ollama LLM (Llama 3.2)**, grounding loan-underwriting responses in user-uploaded documents and a curated bank policy corpus with **metadata-based multi-tenant retrieval scoping**.
- Engineered **structured-extraction prompts** to parse unstructured loan documents into typed **JSON** (income, credit score, DTI, missing docs) via an **LLM**, backed by a deterministic **regex-based fallback** layer to guarantee valid output when generation fails or hallucinates.
- Diagnosed and resolved a **concurrency bug** in the inference layer: synchronous **Ollama** calls were blocking **FastAPI's** single-threaded **event loop**, freezing the app for all users during generation; rearchitected the service to **async/await**, cutting concurrent request latency from 30-60s to under 2s (**95% reduction**).
- Built end-to-end AI-powered banking intelligence features across **20 REST API endpoints** — spanning auth, chat history, and document analysis — chaining **retrieval**, **prompt construction**, and **LLM inference** into a **3-mode analysis engine** (risk assessment, compliance check, executive summary) via **FastAPI** and a **React/TypeScript** frontend.
- Delivered a full-stack, **JWT-secured** platform (1,900 lines backend / 2,600 lines frontend) with multi-turn **chat history**, session persistence, and a **voice-driven query interface (Web Speech API)**, enabling natural interaction with the underlying AI pipeline.

FlightLens – Aviation RAG AI Assistant

- Developed an end-to-end **Retrieval-Augmented Generation (RAG)** application using **Python**, **LangChain**, **FAISS**, **MPNet embeddings**, and **FLAN-T5**, enabling semantic search across **1,000+ aviation documents** and delivering accurate, source-grounded responses for pilots and aviation professionals.

- **Engineered a scalable document ingestion and vector search pipeline** by implementing document chunking, embedding generation, and **FAISS vector indexing**, improving information retrieval relevance by **40%** and reducing query response time to **under 2 seconds**.
- **Built an interactive Streamlit application** with **voice-enabled queries (Whisper)**, live **METAR weather integration**, and optional **Microsoft Flight Simulator telemetry**, enhancing user experience with real-time aviation insights and multimodal interaction.
- **Integrated source attribution and retrieval evaluation** using **LangChain** to provide transparent, citation-backed answers, improving response reliability and reducing hallucinations while demonstrating production-oriented **LLM application** development.

EDUCATION

Master of Science in Data Science	2024-2025
University of North Texas, Denton, TX	GPA: 4.0/4.0
Bachelor of Technology in Electronics and Communication Engineering	2019-2023
Presidency University, Bangalore, India	GPA: 3.2/4.0